

# *Similarity and Diagonalization*

## *Similarity*

**Defn.** Matrices  $A$  and  $B$  are **similar** if there is an invertible matrix  $P$  such that multiplying  $A$  on the one side by  $P$  and on the other side by  $P^{-1}$  gives  $B$ .

Note that the definition is symmetric: if  $B = P^{-1}AP$  then  $A = PBP^{-1}$ .

## *Similarity and Characteristic Polynomial*

**Fact.** *If matrices are similar, then they have the same characteristic polynomial.*

While similar matrices have the same eigenvalues, they usually do not have the same eigenvectors.

## *Full Set of Eigenvectors*

**Defn.** An  $n \times n$  matrix  $A$  is said to have a **full set of eigenvectors** if it has  $n$  linearly independent eigenvectors.

For example, if all eigenvalues are different, then such a full set is guaranteed.

## *Diagonalizable*

**Defn.** A matrix  $A$  is **diagonalizable** if it is similar to a diagonal matrix  $D$ .

If  $D$  exists, its diagonal entries are the eigenvalues of  $A$ .

## *Diagonalization*

### **ALGOR** *Diagonalization*

*If matrix  $A$  has a full set of eigenvectors, then  $A = PDP^{-1}$ , where matrix  $P$  has the eigenvectors of  $A$  as its columns, and diagonal matrix  $D$  has the eigenvalues of  $A$  on its diagonal (in the same order).*

## *Example Diagonalization*

Consider  $A = \begin{bmatrix} 13 & 15 \\ -10 & -12 \end{bmatrix}$ .

Then  $P = \begin{bmatrix} -1 & -3 \\ 1 & 2 \end{bmatrix}$ ,  $D = \begin{bmatrix} -2 & 0 \\ 0 & 3 \end{bmatrix}$ .

(Check!)

## *Diagonalization and Matrix Powers*

**Fact.** If  $A = PDP^{-1}$  then  $A^k = PD^kP^{-1}$ .



## Summary

Matrices  $A$  and  $B$  are similar if there is  $P$  such that  $A = PBP^{-1}$ . One consequence is that  $A^k = PB^kP^{-1}$ . Similar matrices have the same characteristic polynomial and eigenvalues but not eigenvectors.

A matrix similar to a diagonal matrix is diagonalizable. If  $A$  has full set of eigenvectors (e.g. all eigenvalues are different), then  $A = PDP^{-1}$ , where  $P$  has eigenvectors of  $A$  as columns and  $D$  has eigenvalues of  $A$  as diagonal.